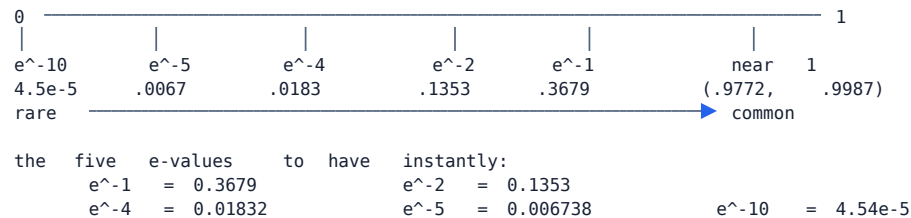
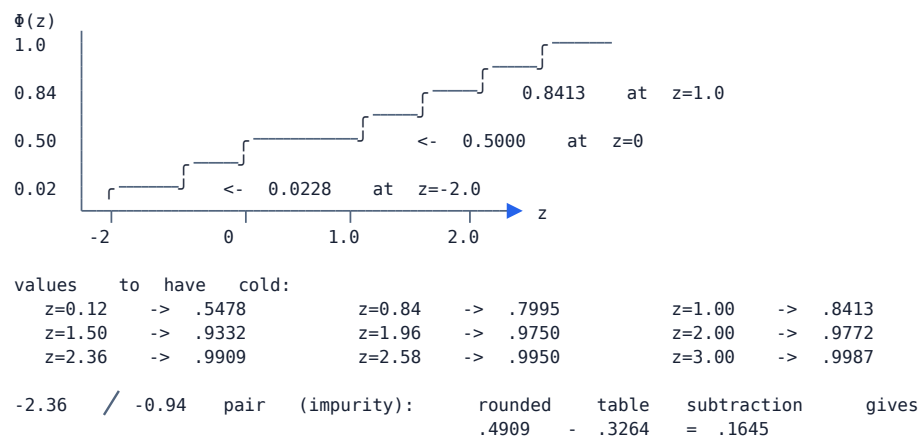


06 NUMBERS: constants, table values, landmarks, identities

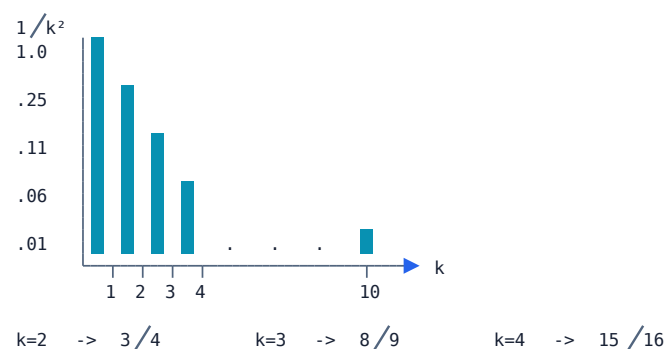
THE NUMBER LINE OF EVERY PROBABILITY YOU WILL WRITE



THE STANDARD NORMAL TABLE AS A PICTURE



THE THREE CHEBYSHEV THRESHOLDS, DRAWN



Every number that can appear, with what it is and where it comes from. Grouped so you can find one fast.

1. Chebyshev thresholds (exact fractions)

k	bound $1/k^2$	complement $1 - 1/k^2$	
1	1	0	(vacuous)
2	$1/4 = 0.25$	$3/4 = 0.75$	
3	$1/9 = 0.1111$	$8/9 = 0.8889$	
4	$1/16 = 0.0625$	$15/16 = 0.9375$	
5	$1/25 = 0.04$	$24/25 = 0.96$	
10	$1/100 = 0.01$	$99/100 = 0.99$	
sqrt(10) approx 3.1623	-> $k^2 = 10$ from a 0.1 tail		

2. Distribution moments worth having instantly

one fair die	$E = 7/2 = 3.5,$	$E(X^2) = 91/6, \text{ Var} = 35/12 = 2.9167$	
sum of two dice	$E = 7,$	$\text{Var} = 35/6 = 5.8333,$	$\text{sd} = 2.4152$
$B(1,p)$ Bernoulli	$E = p,$	$\text{Var} = pq$	
$B(12, 0.1)$	$E = 1.2,$	$\text{Var} = 1.08$	
$B(600, 1/6)$ $E = 100,$	$\text{Var} = 250/3 = 83.3333,$ $\text{sd} = 9.1287$		
$\text{Poi}(2)$	$E = \text{Var} = 2,$	$\text{sd} = 1.4142$	
$\text{Poi}(10)$ $E = \text{Var} = 10$			
$U(2,6)$	$E = 4,$	$\text{Var} = 4/3 = 1.3333$	
source example: mean 8, sd 5. The source writes $N(8,5)$, but its arithmetic treats 5 as			
sigma, not σ^2 . To avoid the ambiguity, write $\mu=8,$ $\sigma=5$, or $N(8,25)$.			
$\text{Exp}(l)$	$E = 1/l,$	$\text{Var} = 1/l^2$	
Exp mean 2	-> $l = 1/2$		

3. Standard normal table values (cumulative, the common ones)

z	Phi(z)	use
0.00	0.5000	centre
0.10	0.5398	
0.12	0.5478	the 8.6 question, $P(X < 8.6)$
0.24	0.5948	
0.50	0.6915	
0.60	0.7257	
0.84	0.7995	
0.8416	0.8000	inverse: a 20 percent lower tail
1.00	0.8413	
1.28	0.8997	the 90 percent landmark
1.34	0.9099	
1.50	0.9332	
1.645	0.9500	95 percent one tail
1.96	0.9750	95 percent two tails
2.00	0.9772	
2.36	0.9909	
2.576	0.9950	99 percent two tails
3.00	0.9987	

negative z, use $\Phi(-z) = 1 - \Phi(z)$, or the table's left tail directly.
the -2.36 / -0.94 pair (impurity): rounded table subtraction gives
 $0.4909 - 0.3264 = 0.1645$ (errata 6)

4. The 68 / 95 / 99.7 landmarks

for a normal distribution:	
within 1 sigma	68.27 percent
within 2 sigma	95.45 percent
within 3 sigma	99.73 percent
(the ppt4 slides round these to 68.3, 95.4, 99.7)	
used by: A2 MCQ Q11, and any "approximately what fraction" question	

5. Worked corpus values (the ones that reappear)

pens B(12,0.1)	$P(X=2) 0.2301$	$P(X \geq 2) 0.3410$	$P(X=0) 0.2824$
irregular die	$p = 5/8$, $P(X=0)$ over 10 = $5.499e-5$, count over 10000 = 0.5499		
insurance Poi(5)	$P(X=4) 0.1755$	(slide 0.1745 is wrong, errata 1)	
calls nesting	$32 e^{-10} = 0.0014528$		
Poisson ratio	$l = 10$, $P(X=0) = 4.54e-5$		
uniform U(2,6)	$P(3 < X < 5) 0.50$		
normal $\mu=8$, $sd=5$	$P(X < 8.6) 0.5478$	$P(X > 8.6) 0.4522$	inverse 20 percent 3.792
two-unknown normal	$\sigma 28.2$, $\mu 37.2$, cutoff 30.4	(errata 10)	
exponential 15/hr	$P(T < 3 \text{ min}) 0.5276$		
exponential mean 2	$P(X < 1 X < 2) 0.6225$	(slide 0.5679 wrong, errata 9)	
exponential lifetime	$P(X > 12)$ with $f = x e^{-x/3} / 9 \rightarrow 5e^{-4} = 0.0916$ (errata 3)		
clt lightbulbs	SE 20 at $n=25$, SE 10 at $n=100$		
clt ATM	$Z = 1.50$, $P = 0.0668$		
clt impurity	$Z1 -2.36$, $Z2 -0.94$. Rounded table areas give 0.1645;		
unrounded z values give about 0.1637	(errata 6)		
clt LED	$Z = -2.00$, $P = 0.0228$		
clt machine $n=9$	$Z1 -1.80$, $Z2 0.60$, answer 0.6898		
estimator set 1	$\lambda 0$, $\text{Var}(t1) 0.2s^2$, $t2 1.5s^2$, $t3 0.5556s^2$, best $t1$		
estimator set 2	$\text{Var}(T1) 3s^2$, $\text{Var}(T2) 29s^2$, $\text{Var}(T3) s^2/3$, best $T3$		
dice Chebyshev	$\sigma^2 35/6$, $k=3$, bound $35/54 = 0.6481$, exact $1/3$		
600 throws	$1 - 5/24 = 19/24 = 0.7917$, exact binomial 0.9754		
marks Chebyshev	$k=2$, $P \geq 0.75$		
tube (errata 16)	$P(X < 150) = 1/3$, $P(X > 150) = 2/3$, all three $1/27$, none $8/27$ (key $2/3$ WRONG)		
hypergeometric lot	mean $nK/N = 4 \times 5/25 = 0.80$ exactly, same as binomial mean		
battery sd	$\text{Var} 0.9475$, exact sd 0.9734 $\rightarrow 0.97$ (key's 0.975 wrong, errata 12)		

6. Identities and algebra shortcuts

$$e^0 = 1$$

$$e^{-1} = 0.3679 \quad e^{-2} = 0.1353 \quad e^{-4} = 0.01832 \quad e^{-5} = 0.006738 \quad e^{-10} = 4.54e-5$$

$$C(n,r) = C(n,n-r)$$

sum $p(x) = 1$ (algebraic check, use it on find-k questions)

35/12 and 35/6 (one die, two dice variance)

$(b-a)^2/12$ uniform variance, 12 is fixed

$\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$ when independent, the minus stays plus

$$\Phi(-z) = 1 - \Phi(z)$$

$$\text{relative efficiency} = \text{Var}(T_2)/\text{Var}(T_1)$$

$$\text{MSE} = \text{Var} + \text{Bias}^2$$

7. The exact-vs-rounded values (write the exact one)

question	write	not
insurance $P(X=4)$	0.1755	0.1745 (slide)
irregular die count	0.5499 $\rightarrow$.55 0.549 (slide)	
assignment Q2 $P(X>12)$	0.0916	0.0915 (key truncates)
battery sd	0.97	0.975 or 0.98 (key rounds twice)
A2 C3 $P(X>2)$	0.8753	0.8754 (key sums rounded terms)
exponential conditional	0.6225	0.5679 (key wrong)
MTE 2024 CDF	option D	option B (scheme wrong)
MTE 2025 Q8(ii)	$E(t^2) \neq \theta^2$	$E(t^2) \neq 0$ (scheme drops θ^2)

8. Unit conversions that decide marks

15 per hour	$\rightarrow$	3 minutes = 0.05 hour	(rate in hours, time in hours)
mean 2 (hours) $\rightarrow$ $\lambda = 1/2$ per hour			
4 minutes mean gap at 15/hour:	$1/\lambda = 1/15$ hour = 4 minutes		
days and hours: convert to ONE unit before substituting			
RULE: the rate and the time MUST share a unit. Convert the odd one first.			